ZEPTO E-COMMERCE PLATFORM

CHAPTER-2

2 Introduction:

Designing a database begins with identifying the key entities, their attributes, primary keys, and the relationships between them. In the Zepto E-Commerce Management System, entities represent the main objects involved in the application, such as Customer, Product, Category, Order, Payment, Delivery, and Admin. Each entity contains attributes that describe its characteristics, while a primary key uniquely identifies each record in the database. Relationships are established between entities to represent how they interact with one another, ensuring efficient data storage, integrity, and retrieval. A well-designed database structure minimizes data redundancy, maintains consistency, and supports the smooth functioning of the e-commerce system.

2.1 Entity Description:

Entity	**Description**
Customer	Stores customer details.
Product	Stores product information.
Category	Stores product categories.
Order	Stores customer order details.
Order Item	Stores products included in each order.
Payment	Stores payment information.
Delivery	Stores delivery details.
Admin	Stores administrator details.

2.2 Customer Table:

Attribute	**Description**
Customer_ID	Unique customer ID
Customer_Name	Customer name
Email	Customer email address

Attribute	Description
Phone_Number	Customer contact number
Address	Customer delivery address
Password	Customer account password

2.3 Product Table:

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Price	Product price
Stock	Available stock quantity
Category_ID	Category ID
Description	Product description

2.4 Category Table:

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name
Description	Category description

2.5 Order Table:

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order

Attribute	Description
Total_Amount	Total order amount
Order_Status	Current order status

2.6Order Item Table:

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Price of the product

2.7 Payment Table:

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	Payment method
Payment_Status	Payment status
Amount	Payment amount

2.8 Delivery Table:

Attribute	Description
Delivery_ID	Unique delivery ID
Order_ID	Order ID

Attribute	Description
Delivery_Partner	Delivery partner name
Delivery_Status	Delivery status
Delivery_Date	Delivery date

2.9 Admin Table:

Attribute	Description
Admin_ID	Unique administrator ID
Admin_Name	Administrator name
Email	Administrator email
Phone_Number	Administrator contact number
Password	Administrator account password

2.10 Primary Keys:

Entity	Primary Key
Customer	Customer_ID
Product	Product_ID
Category	Category_ID
Cart	Cart_ID
Order	Order_ID
Order Item	Order_Item_ID
Payment	Payment_ID
Delivery	Delivery_ID
Admin	Admin_ID

2.11 Relationships Table:

Relationship	**Cardinality**	**Description**
Customer → Cart	One-to-Many (1:M)	One customer can have multiple cart items.
Customer → Order	One-to-Many (1:M)	One customer can place many orders.
Category → Product	One-to-Many (1:M)	One category contains many products.
Product → Cart	One-to-Many (1:M)	One product can be added to many customer carts.
Order → Order Item	One-to-Many (1:M)	One order can contain multiple products.
Product → Order Item	One-to-Many (1:M)	One product can appear in multiple order items.
Order → Payment	One-to-One (1:1)	Each order has one payment record.
Order → Delivery	One-to-One (1:1)	Each order has one delivery record.
Admin → Product	One-to-Many (1:M)	One admin manages multiple products.
Admin → Order	One-to-Many (1:M)	One admin manages multiple orders.

2.12Conclusion:

The identification of entities, attributes, primary keys, and relationships provides a strong foundation for the **Zepto E-Commerce Management System**. It ensures organized data storage, maintains data integrity, reduces redundancy, and enables efficient management of customers, products, orders, payments, and deliveries. This database design supports the smooth and reliable functioning of the system.